The 40th ACM/SIGAPP Symposium On Applied Computing

HARIN: A Novel Metric for Hierarchical Topic Model Assessment

SAC 2025




Harin Seo **Hwanseong Joo** Byungchul Tak Young-Kyoon Suh

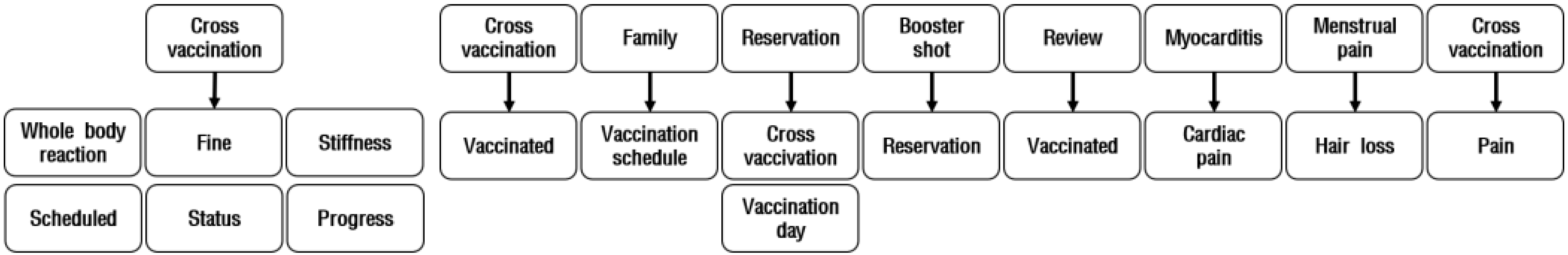
2025. 04. 01

Big Data & AI Convergence System Laboratory
Kyungpook National University

Contents

Introduction: Background

- <u>H</u>ierarchical <u>T</u>opic <u>M</u>odeling (HTM)
  - Popular technique used to **discover significant topics** within datasets and **visualize** the outcome in a **tree-like structure**.
- Previous Research
  - Previous research [1-4] used **hierarchical topic modeling** to gain valuable insights from social media datasets.
    - Ex. Our prior work [5] utilized hierarchical topic modeling to explore opinions regarding COVID-19 discourse from twitter(now 'X') datasets.



Introduction: Motivation

- Drawbacks of Existing HTMs:
  - Generation of **excessive** number of topics, leading to **difficulty in interpretation**.

- Drawbacks of the Coherence score used as a popular HTM Evaluation Metric:
  - Coherence is based on Normalized Pointwise Mutual Information (NPMI).
    - But the NPMI considers only intra-topic but *not* inter-topic relationships.
  - Coherence tends to decrease as topics increase.
    - This tendency of the coherence decrease poses the following challenges:
      1. Complexity of the produced HTMs,
      2. Diluted semantic content, and
      3. Reduced interpretability.

Overcoming these drawbacks solicits a new HTM evaluation metric!

Related Work

Introduction: Contribution

- This paper proposes a novel HTM evaluation metric, HierArchical haRmony INdex (**HARIN**).
  - HARIN is designed to evaluate the quality of hierarchical topic models reflecting the structural characteristics of topics.
  - HARIN integrates the **horizontal and vertical structural characteristics** in a given topic hierarchy.
  - HARIN captures three characteristics of hierarchically organized topics in a holistic fashion:
    - ***Diversity*** between sibling topics,
    - ***Similarity*** between parent and children topics, and
    - ***Coherence*** of the whole topic.

Related Work

- Existing Topic Modeling Evaluation Metrics:

Metric	Limitation	Reference
Hierarchical Affinity	**Limited reflection of complex hierarchical connections**	Shahid et al. [9]
HCoherence	**Does not fully capture hierarchical structures**	Shahid et al. [9]
Word Frequency Similarity	**Considers only parent-child relationships**	Lindgren [12]

- These existing metrics merely considered vertical *or* horizontal relationships of a produced topic hierarchy rather than a single harmonized metric.

Problem Statement

Problem Statement

- This paper's objective:
  - To define a novel metric, M, to comprehensively evaluate the overall quality of disparate HTMs (using *diversity*, *similarity*, and *coherence*).
    - M should be the most suitable scoring metric that can assess the overall quality of the topic hierarchy produced by an HTM.
  - The performance of M is evaluated by **its proximity to human-generated ratings** and **the number of leaf topics after pruning**.

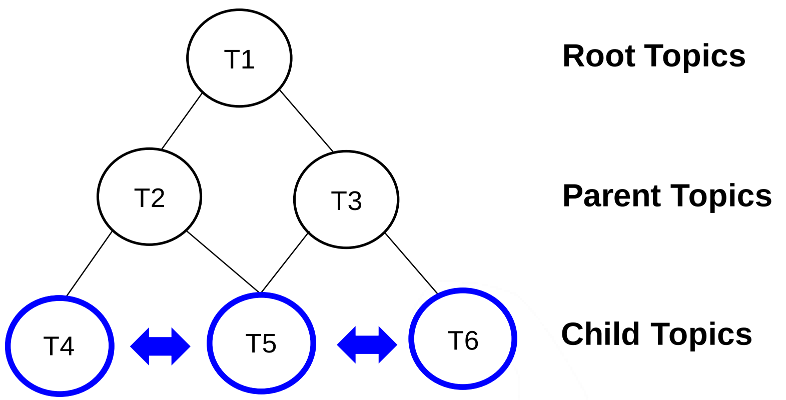
Our Approach (1 / 6): HARIN

- HARIN is defined as $\boxed{\textbf{HARIN} = w_d D_H + w_s S_H + w_c C}$
  - Diversity (D_H)
    - Diversity between sibling topics based on word vector distances.
  - Similarity (S_H)
    - Uses Word2Vec embeddings to calculate cosine similarity between parent and child topics.
  - Coherence (C)
    - Uses NPMI (Normalized Pointwise Mutual Information) to assess the semantic relationship of topics.
  - Notes
    - The sum of the weights ($w_d + w_s + w_c$) is equal to 1.
    - Set depth 3 to ensure fair comparison with other models.
- Interpretation
  - The higher the HARIN score, the better the quality of a given hierarchical topic model.

Our Approach (2 / 6)

- **Diversity** based word embedding:

$$D(T_1, T_2) = 1 - S(T_1, T_2)$$



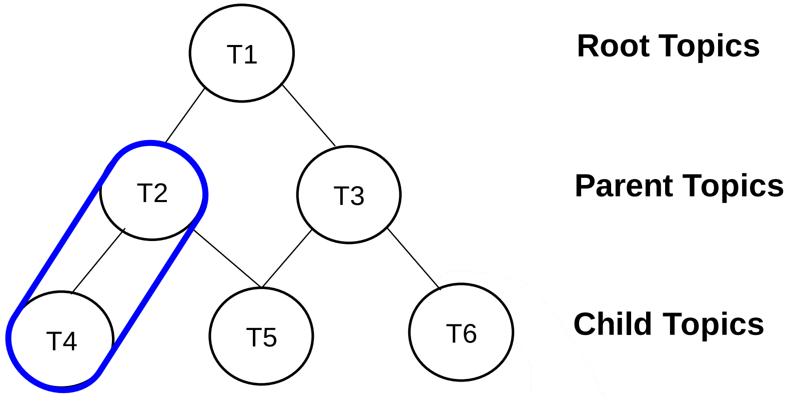
  - Measures how diverse the sibling topics are, or the degree of diversity between sibling topics.
  - Calculates based on word embedding-based distance.
  - Derives D by examining how many various sibling topics exist.

T_1, T_2 : sibling topics

Our Approach (3 / 6)

- **Similarity-based** word embedding:

$$S(T_1, T_2) = \frac{\sum_{i=1}^{n} \sum_{j=1}^{m} \mathrm{Sim}(T_{1i}, T_{1j})}{n \times m},$$



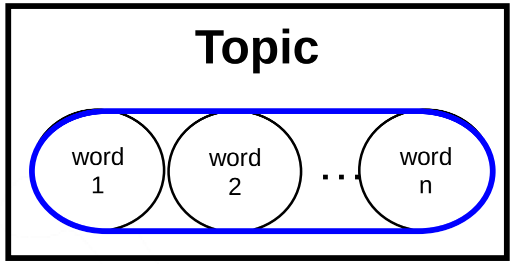
- Measures how similar the parent topic is to its child topic or the similarity between parent-child topics.
- Calculates word-embedding-based similarity between T_1's i-th and T_2's j-th words.
- Penalty halves the derived similarity if more than half of the words in two topics between parent and child are the same.

T_{1i}: the i-th word vector in T_1
T_{2j}: the j-th word vector in T_2
n, m: # of top-k words in topic T_1 and T_2

Our Approach (4 / 6)

- **Topic Coherence**:

$$\mathrm{NPMI}(o_i, o_j) = -\frac{\left(\log \frac{p(o_i,o_j)}{p(o_i)p(o_j)}\right)}{\log p(o_i, o_j)} \qquad \mathrm{C}(o_i,o_j) = \frac{\sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \mathrm{NPMI}(o_i, o_j),}{\binom{n}{2}}$$



  - Normalized PMI(NPMI) is used to assess the semantic coherence of a topic by measuring the strength of association between the words *within* that topic.
  - Higher NPMI values indicate that the words in a topic are semantically related: thus, a more coherent topic.
  - Coherence score is *normalized* between 0 and 1.
    - Same as those of similarity and diversity.

o_i, o_j: the *i*-th and *j*-th words belonging to the topic
$\binom{n}{2}$: total number of word pairs

Our Approach (5 / 6)

- Calculation of HARIN Score:

  1. `CalcDiversity` Function
     - Computes the diversity of a child nodes based on top-k words.
     - Iterates over all combinations to calculate diversity.
     - Averages the diversity score across all combinations to obtain the final value.
  2. `CalcSimilarity` Function
     - Calculates the similarity score to each child node, accumulates the results, and returns the similarity score.
  3. `CalcTotalScore` Function
     - Calculates diversity, similarity, and coherence scores, then combines them via a weighted sum to produce the final HARIN score.

Algorithm 1: Calculating the Hierarchical Harmony Index (HARIN) Score on a Given Hierarchical Topic Model

```
Input  : A given hierarchical topic model (m), the number of top
         words (k) belonging to each node in the topic hierarchy
         (H) produced by m, and a weight vector W = [w_s, w_d, w_c]
Output : Calculated the HARIN score (score)
Function CalcDiversity(node, k):
    D ← 0;
    child_list ← node.childs;
    for i ← 0 to len(child_list) − 1 do
        for j ← 0 to i − 1 do
            D += (1 -
             getSimilarity(child_list[i], child_list[j], k));
             // Apply Eq. 3.
        end
    end
    D /= Σ_{i=0}^{len(child_list)−1} i;
    return D;
End Function
Function CalcSimilarity (node, k):
    S ← 0;
    foreach c ∈ node.childs do
        S + = getSimilarity(node, child, k) ;        // Apply Eq. 2.
    end
    S /= |node.childs|;
    return sim_s;
End Function
Function CalcTotalScore (m, k, W):
    H ← Obtain a topic hierarchy produced by m.
    D_H ← 0;
    S_H ← 0;
    C ← H.get_coherence();
    for l ← 0 to H.level do
        foreach node ∈ H.nodes(level) do
            if node.childs == NULL then continue ;
            D_H += CalcDiversity(node, k);
            S_H += CalcSimilarity(node, k);
        end
    end
    D_H /= |H.nodes| − |H.leaf|;
    S_H /= |H.nodes| − |H.leaf|;
    return (w_d×D_H+w_s×S_H+w_c×C) ;               // Apply Eq. 1.
End Function
```

Our Approach (6 / 6)

- Selection of the Best-fitting model

1. Preprocessing
   - Preprocess the dataset (*RD*).
2. Initialization
   - Set *best_mdl* and *best_score* to initial values.
3. Model Training
   - For each model (*m*) in the list (*L*), train the model using the preprocessed dataset.
   - Calculate the HARIN score for the trained model (*tm)* based on the top-k words in each topic.
4. Score Comparison
   - Update *best_mdl* and *best_score* if HARIN score exceeds the current *best_score*.

Algorithm 2: Selection of the Best-fitting Model

```
Input  : A given raw dataset (RD), a list (L) of candidate
         hierarchical topic models to be trained using RD, and the
         number (k) of top words to consider in a topic
Output : The selected hierarchical topic model (best_mdl)
PD ← Preprocessing(RD);
best_mdl ← None;
best_score ← −1;
foreach m ∈ L do
    tm ← m.train(PD);
    score ← CalcTotalScore(tm, k) ;        // See Algo. 1.
    if best_score < score then
        best_mdl ← tm;
        best_score ← score;
    end
end
return best_mdl;
```

Experimental Design

Experimental Design (1 / 3)

- Research Questions

1. How *consistently* does HARIN choose an optimal hierarchical topic model across various real-world datasets?
2. How *similarly* does HARIN rank competing hierarchical topic models as the human evaluators do, compared to other baseline measures?
3. How *significantly* does each component of HARIN contribute to the performance?
4. How *effectively* does HARIN prune leaf topics and thus enhance interpretability?

Experimental Design (2 / 3)

- Datasets

Datasets	Description	# of records	# of words
COVID-19 (ours)	Korean tweets on vaccine brand (February 2021 ~ July 2021)	29,297	169,031
20 Newsgroups	Newgroups posts on 20 topics	18,846	278,723
PubMed	Abstract on Type 2 Diabetes Mellitus	9,211	1,065,818
Yelp	Korean food discourse	28,516	1,278,126
arXiv	Abstract from arXiv papers regarding Computer Science	7,945	766,232

Experimental Design (3 / 3)

- Hierarchical Topic Models:

Model Type	Models	Characteristics	Reference
LDA-Based	hLDA, KeyETM	Probabilistic hierarchy (nCRP), Topic-word distribution	Thomas et al. [6] Harandizadeh et al. [7]
NMF-Based	CluHTM, HyHTM	Non-negative matrix factorization, Hyperbolic geometry	Felipe Viegas et al. [8] Shahid et al. [9]
Embedding-Based	BERTopic, Top2Vec	Pretrained contextual embeddings	Grootendorst [10] Dimo Angelov [11]

- To validate HARIN metric, this paper considers 4 representative hierarchical topic models chosen from each category (BERTopic, hLDA, CluHTM, and HyHTM) for evaluation.

Evaluation Results (1 / 5)

- **Optimal Model Choice Consistency Evaluation** (to respond to RQ1):
  - (RQ1) "How *consistently* does HARIN choose an optimal hierarchical topic model across various real-world datasets?"

Datasets	COVID-19					20Newsgroups					PubMed					Yelp					arXiv				
Models / Metrics	Sim.	Div.	Coh.	HRN	HUM	Sim.	Div.	Coh.	HRN	HUM	Sim.	Div.	Coh.	HRN	HUM	Sim.	Div.	Coh.	HRN	HUM	Sim.	Div.	Coh.	HRN	HUM
BERTopic	0.61	0.46	0.35	0.47	0.66	0.43	0.58	0.48	0.49	0.56	0.19	0.84	0.51	0.51	0.69	0.17	0.82	0.51	0.50	0.64	0.23	0.78	0.50	0.49	0.68
CluHTM	0.23	0.41	0.70	0.44	0.56	0.58	0.49	0.44	0.50	0.70	0.39	0.71	0.38	0.49	0.59	0.29	0.73	0.40	0.47	0.48	0.46	0.57	0.35	0.45	0.60
hLDA	0.54	0.58	0.53	**0.55**	**0.79**	0.55	0.51	0.55	**0.53**	**0.81**	0.20	0.83	0.53	**0.52**	**0.88**	0.17	0.87	0.52	**0.51**	**0.79**	0.21	0.77	0.51	**0.50**	**0.73**
HyHTM	N/A					0.52	0.47	0.40	0.46	0.53	0.41	0.57	0.37	0.44	0.45	0.35	0.81	0.37	0.51	0.36	0.44	0.52	0.35	0.43	0.43

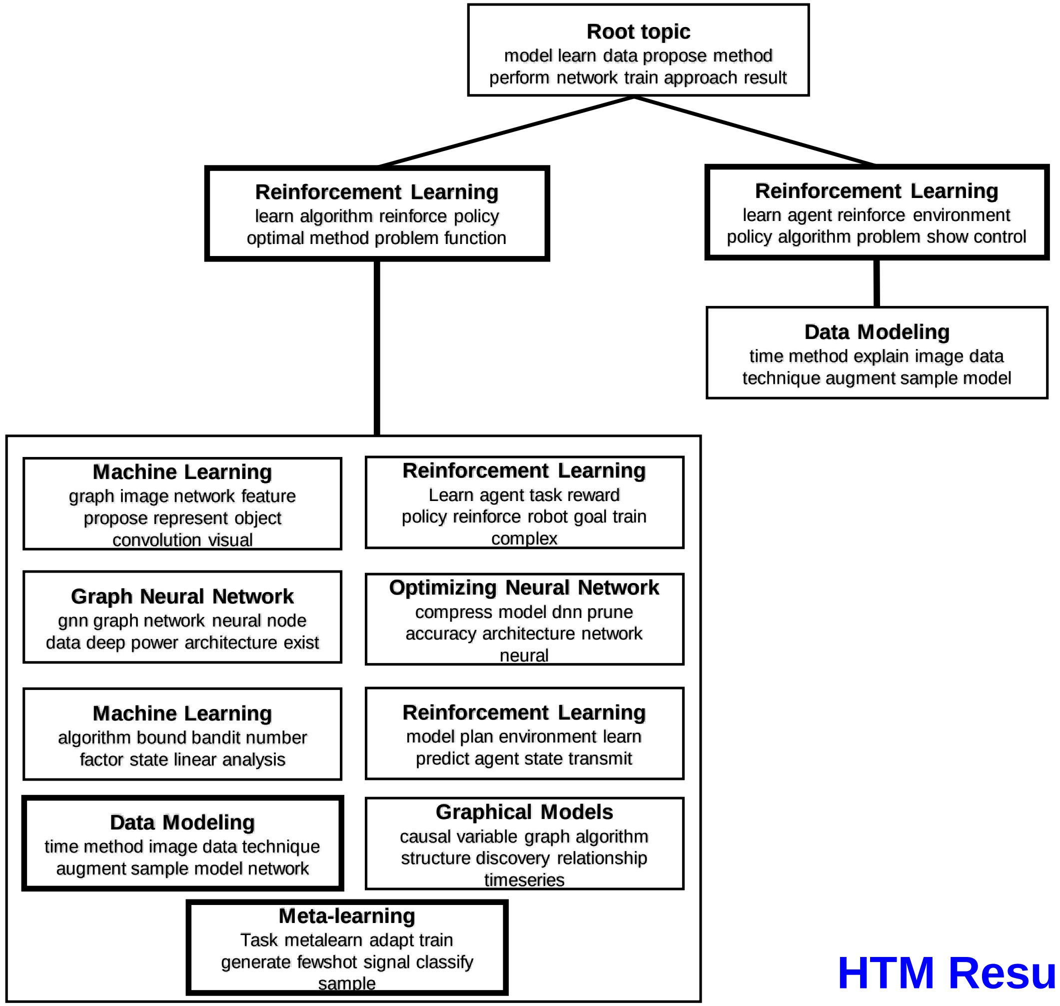
Sim. : Similarity, **Div.** : Diversity, **Coh.** : Coherence, **HRN** : HARIN, **HUM** : Human Score

- Overall, **hLDA** achieved the highest HARIN score due to balanced scores across metrics.
- **hLDA** applying HARIN produces a well-balanced topic hierarchy without much information loss at leaf levels.

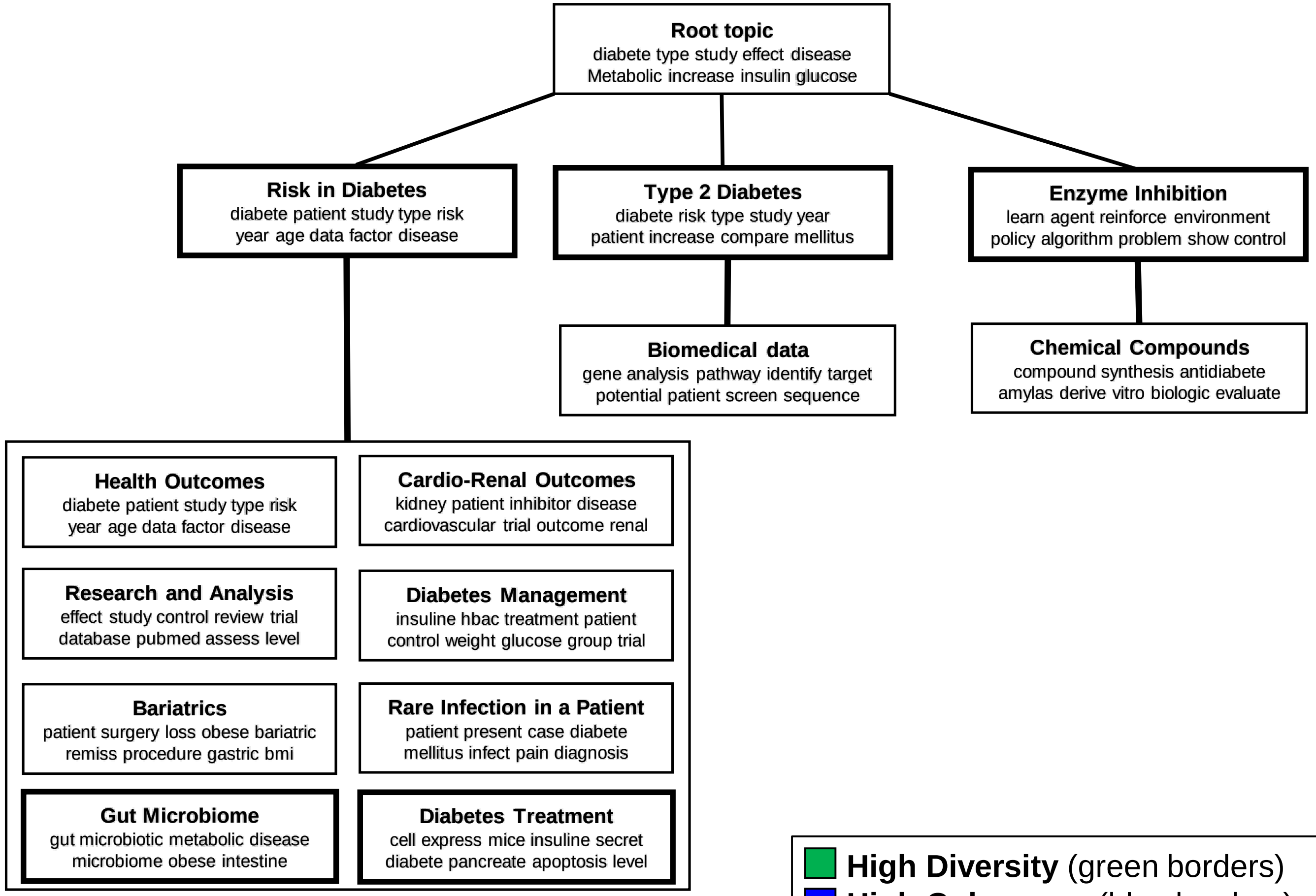
Evaluation Results (2 / 5)

- **Optimal Model Choice Consistency Evaluation** (to respond to RQ1) (Cont'd):

Ex 1) arXiv Dataset



Ex 2) PubMed Dataset



HTM Results by hLDA applying HARIN

High Diversity (green borders)
High Coherence (blue borders)
High Similarity (orange lines)

Evaluation Results (3 / 5)

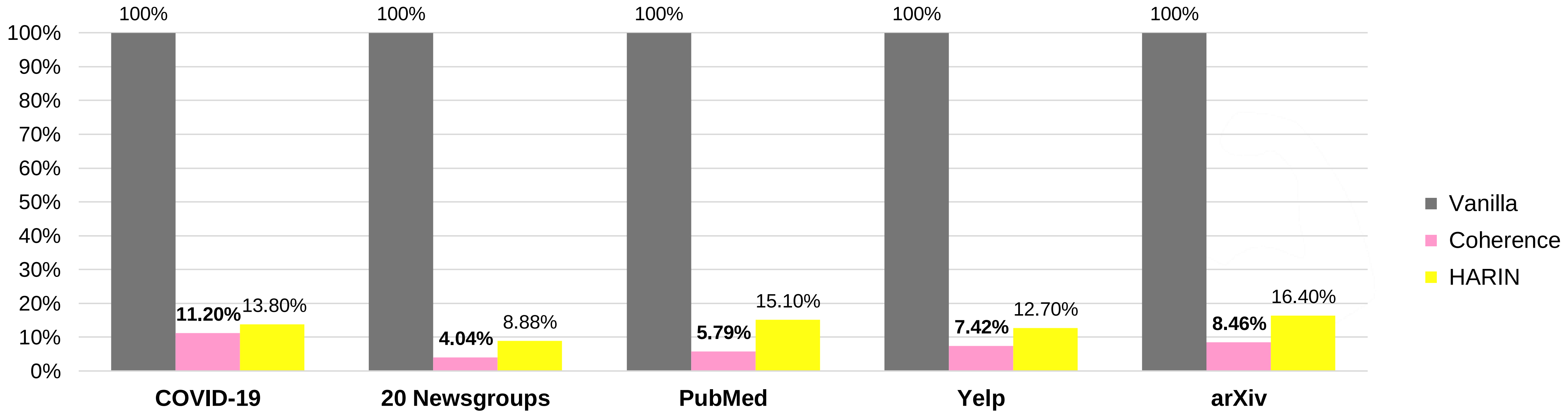
- Ranking Alignment Evaluation (to answer RQ2):
  - (RQ2) "How *similarly* does HARIN rank competing hierarchical topic models as the human evaluators do, compared to other baseline measures?"

Dataset	Metrics						
	Sim.	Div.	Coh.	Sim.+Div.	Div.+Coh.	Sim.+Coh.	HARIN
COVID-19	0.33	1	0	1	0.33	1	1
20 Newsgroups	0	0.25	0.5	0.5	0.25	0.25	1
PubMed	0	0.25	1	0.25	1	0	1
Yelp	0	0.5	0.5	0.25	0.5	0	**0.25**
arXiv	0	0.5	1	0.5	1	0	1
Average	0.07	0.55	0.60	0.50	0.62	0.25	**0.85**
Mean Reciprocal Rank (MRR)	0.37	0.67	0.87	0.48	0.87	0.58	**1**

  - HARIN's rankings on average 85% were consistent with the human score based rankings across the datasets.
  - HyHTM was ranked fourth by human evaluators on the Yelp dataset resulting in a ranking accuracy 0.25.
  - Coherence metric alone is *insufficient* since humans consider similarity and diversity in topic evaluation.

Evaluation Results (4 / 5)

- Evaluation of Effectiveness of Pruning and Enhancing Interpretability (to respond to RQ4)
  - (RQ4) “How *effectively* does HARIN prune leaf topics and thus enhance interpretability?”

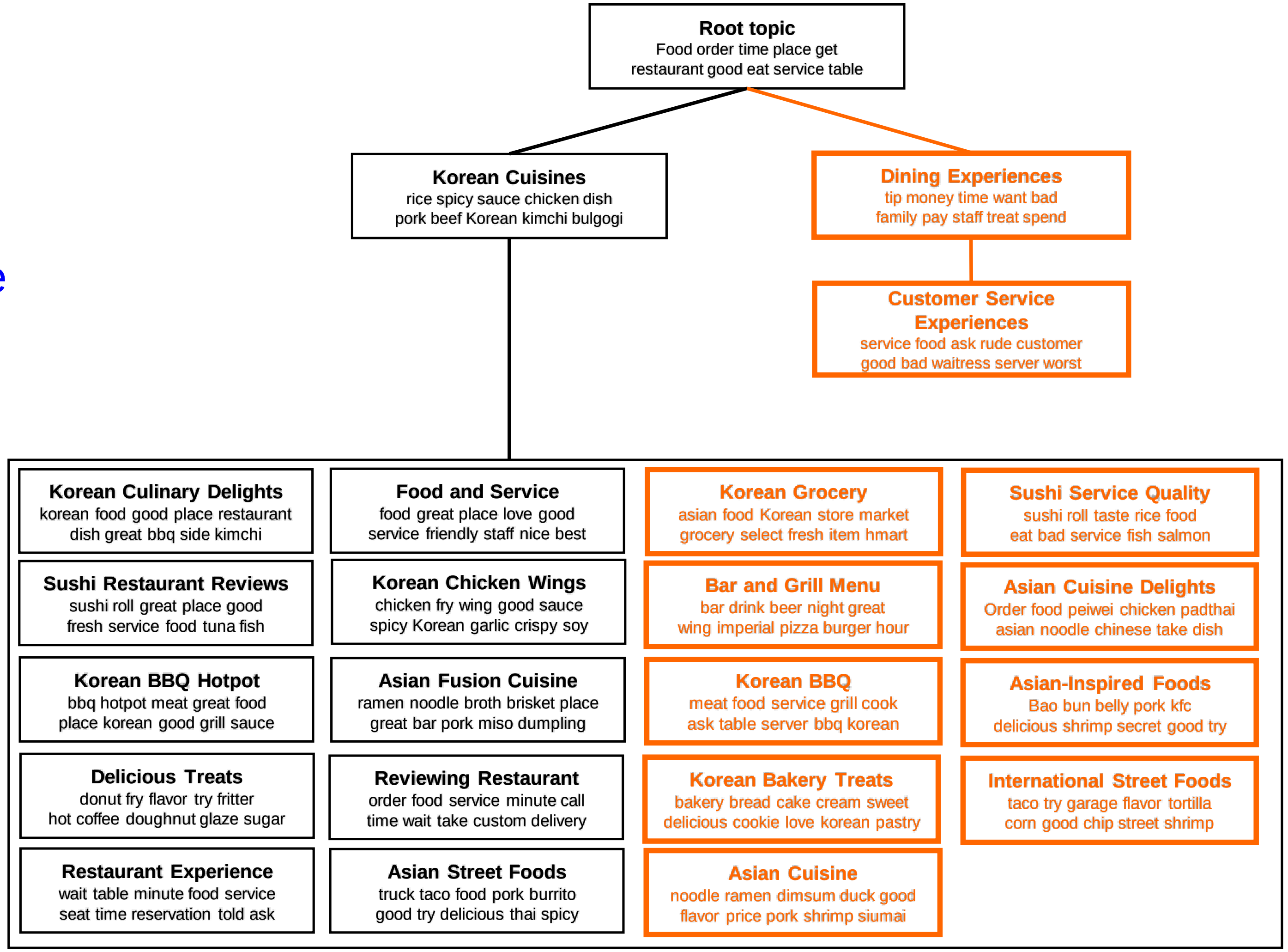


Percentage comparison of leaf topics remaining after pruning by each metric:
Vanilla (without pruning leaf topics) vs. Coherence vs. **HARIN**

- Vanilla refers to baseline without pruning leaf topics, showing 100% retention.
- Coherence metric prunes average approximately 5.99% *more* leaf topics than HARIN.
- HARIN prunes *fewer* topics while maintaining both accuracy and interpretability.

Evaluation Results (5 / 5)

- Evaluation of Effectiveness of Pruning and Enhancing Interpretability (to respond to RQ4)
  - Coherence often overprunes, potentially leading to *"information loss."*
  - **HARIN** preserves more meaningful topics and provides a more accurate and diverse representation.
    - **Coherence** simplifies **broader categories**: e.g., *Korean foods*
    - **HARIN** captures **more granular topics**: e.g., *Korean Grocery*



※ **Black** nodes indicate topics retained by Coherence.
※ **Orange** + **Black** nodes indicate topics retained by HARIN.

Summary

Conclusion

- We proposed a novel metric, HARIN, to address the challenge of evaluating the interpretability of hierarchical topic models on diverse datasets.
- We validated several real-world datasets to determine the most suitable model for each dataset.
- As a result, hLDA was consistently chosen by HARIN as the best HTM across the datasets and results were consistent with those of human evaluation.
- HARIN can effectively determine the best hierarchical topic model on a dataset of interest without human intervention.
- HARIN enhances interpretability with minimal pruning of meaningful leaf topics.

Future Works

- Developing an algorithm to determine the optimal depth of topic hierarchy in top-down methodologies.
- Exploring a broader range of models, such as hierarchical topic graphs and large language-based topic models.
- Expanding the analysis to include social media texts from emerging platforms and enhance the generalizability of findings.
- Developing techniques for dynamically assigning weights to each component of HARIN, customized to the unique characteristics of datasets.

THANK YOU

Code

https://github.com/lab-paper-code/covaxx_htm

Email:

joohs0505@knu.ac.kr

References

[1] Abdulqader Almars et al., “Modelling User Attitudes using Hierarchical Sentiment-Topic Model,” *Data & Knowledge Engineering*, Vol. 119, pp. 139–149, 2019.

[2] Hyeju Jang, Emily Rempel, David Roth, Giuseppe Carenini, and Naveed Zafar Janjua, “Tracking COVID-19 Discourse on Twitter in North America: Infodemiology Study Using Topic Modeling and Aspect-based Sentiment Analysis,” *Journal of Medical Internet Research*, Vol. 23, No. 2, e25431, 2021.

[3] Aditya Joshi, Pushpak Bhattacharyya, and Mark Carman, “Political Issue Extraction Model: A Novel Hierarchical Topic Model that Uses Tweets by Political and Non-political Authors,” *Proceedings of the 7th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis,* pp. 82–90, 2016.

[4] Sungkyu Park, Sungwon Han, Jeongwook Kim, Mir Majid Molaie, Hoang Dieu Vu, Karandeep Singh, Jiyoung Han, Wonjae Lee, and Meeyoung Cha, “COVID-19 Discourse on Twitter in Four Asian countries: Case Study of Risk Communication,” *Journal of Medical Internet Research,* Vol. 23, No. 3, e23272, 2021.

[5] Harin Seo, Hwanseong Joo, and Young-Kyoon Suh, “A Hierarchical Topic Modeling Analysis of Twitter Data on COVID-19 Vaccines,” *Proceedings of the 2024 ACM International Conference on Research in Adaptive and Convergent Systems* (RACS’24), pp. 149-152, 2024.

[6] Thomas Griffiths, Michael Jordan, Joshua Tenenbaum, and David Blei, “Hierarchical Topic Models and the Nested Chinese Restaurant Process,” *Advances in Neural Information Processing Systems* 16, pp. 17–24, 2003.

[7] Bahareh Harandizadeh, J. Hunter Priniski, and Fred Morstatter, “Keyword Assisted Embedded Topic Model,” *Proceedings of the 15th ACM International Conference on Web Search and Data Mining* (WSDM ’22), pp. 372–380, 2022.

[8] Felipe Viegas et al., “CluHTM - Semantic Hierarchical Topic Modeling based on CluWords,” *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (ACL’20), pp. 8138–8150, 2020.

[9] Simra Shahid et al, “HyHTM: Hyperbolic Geometry-based Hierarchical Topic Model,” *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics* (ACL’23), pp.11672–11688, 2023.

[10] Maarten Grootendorst, “BERTopic: Neural Topic Modeling with a Class based TF-IDF Procedure,” arXiv preprint arXiv:2203.05794, 2022.

[11] Dimo Angelov, “Top2Vec: Distributed Representations of Topics,” arXiv:2008.09470 [cs.CL], 2020.

[12] Jennifer Lindgren, Evaluating Hierarchical LDA Topic Models for Article Categorization, Master’s thesis, Department of Computer and Information Science, Linköping University, Sweden.

Appendix

Human Evaluation Example

- Example of Human Evaluation

Hierarchical Topic Model (HTM) 1
You may find that some words are truncated in the results below. That is because we used a rule-based tokenizer. L0 is the root topic, L1 indicates L0's child, and L2 is L1's child.

L0 : Dining

- ['food', 'korean', 'place', 'order', 'chicken', 'time', 'great', 'service', 'restaurant', 'spicy', 'dish', 'back', 'love', 'eat', 'fry', 'sushi', 'rice', 'delicious', 'meat', 'sauce']

|===**L1 : Korean Cuisine**

- ['food', 'korean', 'place', 'order', 'time', 'great', 'chicken', 'service', 'restaurant', 'spicy', 'dish', 'back', 'love', 'eat', 'meat', 'rice', 'bbq', 'sauce', 'fry', 'delicious']

|======**L2 : Korean Cuisine**

- ['food', 'korean', 'place', 'order', 'time', 'great', 'service', 'restaurant', 'chicken', 'dish', 'spicy', 'back', 'love', 'meat', 'eat', 'bbq', 'rice', 'delicious', 'sauce', 'side']

|======**L2 : Korean Fried Chicken**

- ['wing', 'chicken', 'order', 'bonchon', 'fry', 'spicy', 'place', 'food', 'time', 'garlic', 'soy', 'sauc', 'great', 'wait', 'flavor', 'service', 'crispi', 'back', 'korean', 'half']

|===**L1 : Culinary Delights**

- ['donut', 'sushi', 'chicken', 'place', 'food', 'great', 'order', 'korean', 'time', 'fry', 'fresh', 'flavor', 'love', 'service', 'bowl', 'delicious', 'roll', 'back', 'hot', 'restaurant']

|======**L2 : Flavorful Dining**

- ['donut', 'chicken', 'place', 'food', 'sushi', 'korean', 'order', 'fry', 'great', 'flavor', 'time', 'bowl', 'hot', 'fresh', 'delicious', 'love', 'feder', 'back', 'taste', 'rice']

|======**L2 : Delicious Treats**

- ['donut', 'feder', 'coffee', 'fancy', 'doughnut', 'chicken', 'cinnamon', 'lavend', 'glaze', 'sugar', 'chocol', 'morn', 'vanilla', 'indian', 'fry', 'dozen', 'strawberri', 'earli', 'flavor', 'hot', 'honey', 'dri', 'buttermilk', 'counter', 'hype', 'spice', 'ranch', 'half', 'zaatar', 'rub']

1-1. Do you think that a word included in each topic is correctly assigned to that topic? *

- Absolutely not
- Partially
- Yes

1-2. Do you think each child's topic is similar to its parent's topic? *

- Absolutely not
- Partially
- Yes

1-3. Do you think child topics under the same parent are diverse? *

- Absolutely not
- Partially
- Yes

1-4. Do you think parent and child topic words are semantically coherent? *

- Absolutely not
- Partially
- Yes

Exampl
e

Questio
n

Configuration Settings

Category	Item	Specifications
Hardware	OS	Ubuntu 18.04.1 LTS
	CPU	Intel Xeon E5-2698
	GPU	Tesla V100-DGXS-32GB (x4)
	Memory (RAM)	256GB
	Storage (SSD)	7.2TB (1.8TB x 4)
Software	Programming Language	Python 3.8.19
	Preprocessing Libraries	Konlpy 0.6.0, Gemsim 4.1.2
	Hierarchical Topic Modeling	Tomotopy 0.12, BERTopic 0.14.1

Ranking Alignment

- Ranking Alignment metrics
  - Ranking Accuracy

$$\text{Accuracy} = \frac{1}{|D|}\sum_{i=1}^{D}\frac{1}{|M|}\sum_{j=1}^{M} f(\text{rank}_{ij}^{e}, \text{rank}_{ij}^{H}),$$

D : Number of datasets
M : Number of hierarchical topic models
rank_{ij}^{e} : Rank of the j-th model on the i-th dataset by metric e
rank_{ij}^{H} : Rank of the j-th model on the i-th dataset by human evaluation

    - Measures how well a **given metric (*e*) aligns with human rankings (*H*).**

  - Mean Reciprocal Rank(MRR)

$$MRR = \frac{1}{|M|}\sum_{m=1}^{|M|}\frac{1}{\text{rank}_m},$$

M : Number of hierarchical topic models
rank_m : Rank of the m-th model based on the evaluation metric

    - MRR value is 1, with higher values indicating better performance.
    - Evaluates ranking quality, with **more weight on the top-ranked models.**